

About me

Freshman at Texas A&M University - Prospective aerospace engineering major

I enjoy part design and manufacturing, across both professional and personal projects. I also have strong interests in spaceflight, simulation analysis, and CAD modeling.

At Lockheed Martin I worked defect resolution on a large flight software program, comparing written requirements against implementation. Through NCSSE's Student Spaceflight Experiment Program (SSEP), I helped write and design a germination experiment to be tested in a microgravity environment in the ISS.



LOCKHEED MARTIN



Skills



GAZEBO



CAD

Inventor · SolidWorks · Fusion · Onshape

Manufacturing

CNC milling · CNC routing · manual machining

Simulation

SolidWorks · ROS · URDF · Gazebo

Tools

Git · Gitlab · JIRA

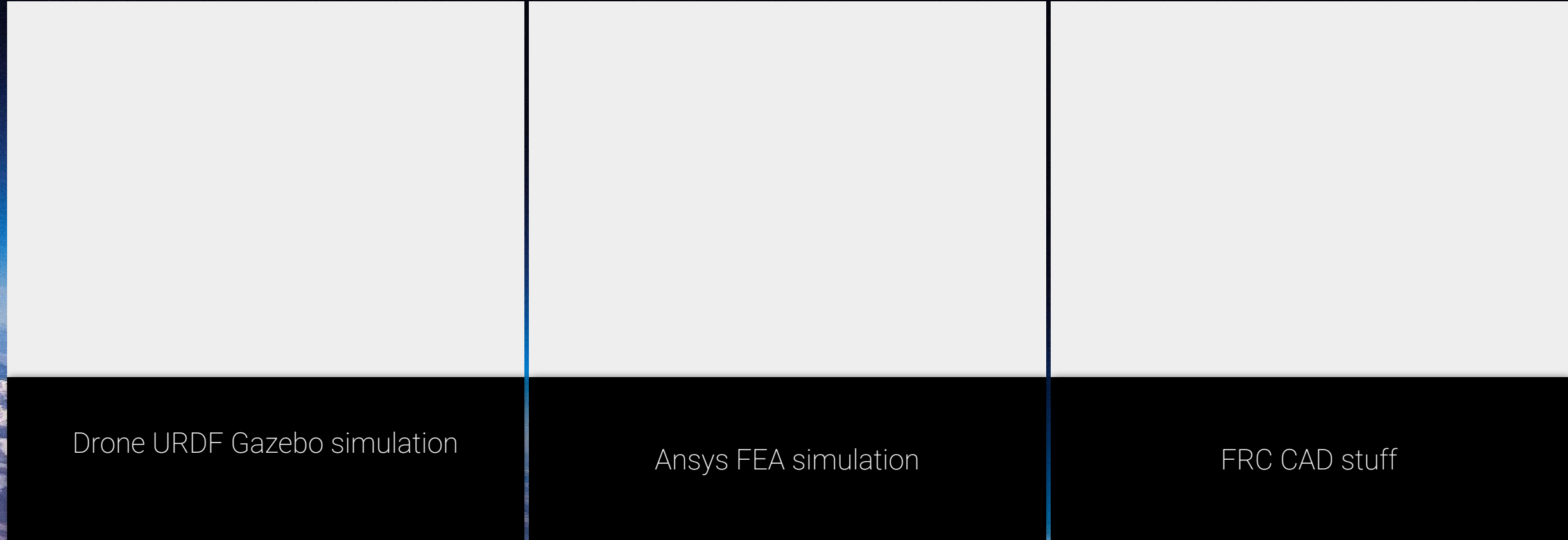
Contact Info

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Drone URDF Gazebo simulation

Ansys FEA simulation

FRC CAD stuff



NCESSE Mission 18 to ISS
Student Researcher

Arlington, TX
Aug, 2023 - Nov, 2024



Lockheed Martin
F-35 System Design Intern

Fort Worth, TX
Sept, 2025 - Aug, 2026

SSEP Mission 18 - Student Researcher

Germination of *Pisum sativum* in microgravity sent to ISS on SpaceX CRS-31 for a resupply mission

Objective: Determine whether pea shoots germinate in microgravity as well as they do on Earth, as a step toward evaluating them as a low maintenance food source that's sustainable on long duration missions. A successful experiment meant that a sealed, unpowered experiment that could run itself for the length of the mission and return a sample comparable to a matched ground control.

Constraints:

- Sealed silicone tube (FME), three compartments, two clamps
- No power, stored in complete darkness
- Two crew interactions total, no other access after launch
- Identical ground control run in Arlington

Sequence:

- Arrival: first clamp opened, 3 mL sterile water starts germination
- Undock: 2 days, second clamp opened, preservative stops growth before reentry

Iterations:

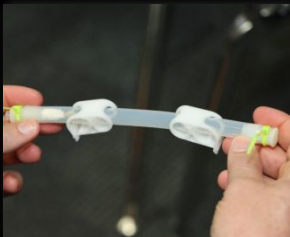
The first configuration used three specimens, 3 mL of water across two compartments, and a 16 in² towel. Over a 2 week trial the peas rotted instead of sprouting because of too much water and little room for germination. Reduced to one pea, less water, 1 in² towel.

Germination in darkness was unproven, so we created a simple 3D-printed box with a sliding door and ran a tube inside it. The roots and shoots developed, so we moved the specimen to the end compartment for shoot clearance above and root space below.

Result:

Flight sample germinated more successfully than the ground control, indicating that microgravity did not impair germination under these conditions.

Silicone Fluid Mixing Enclosure (FME)



Compartment	Contents	Volume	Released
1	One dried pea, 1 in² paper towel	0.3 mL (pea)	-
2	Sterile water (DEPC-treated)	3 mL	Clamp A, on arrival
3	Preservative (10% neutral buffered formalin)	3 mL	Clamp B, undock -2 days

Preliminary germination trials (2 week period)



Links:

[Flight safety full documentation](#)

[NCESSE SSEP Mission 18 Overview](#)

[NCESSE SSEP - Arlington, TX](#)

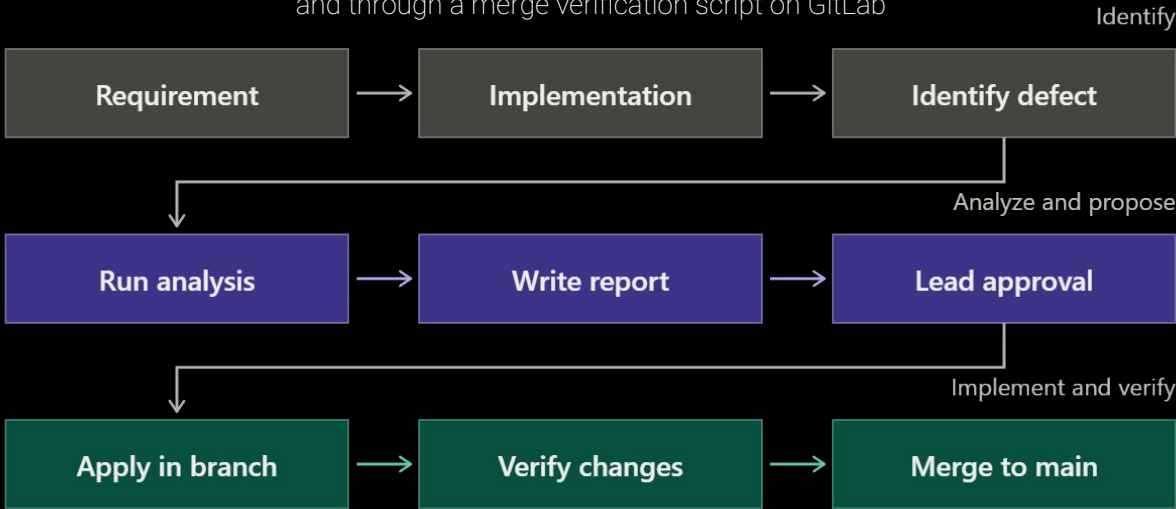
[Martin High School - STEM Academy](#)

Program Partners

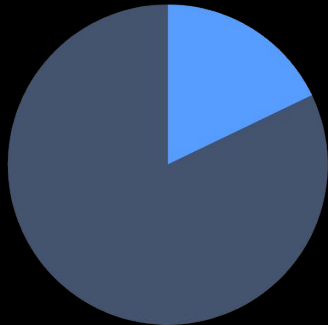


Defect Resolution Workflow

Work is verified through a software team lead for the related technical area and through a merge verification script on GitLab



Technical Debt Reduction



Status	Issues	%
Resolved	~157	~18
Remaining	~722	~82
Total	~879	100

F-35 System Design Intern



Role:
Resolve defects in system level requirements (the layer under flight software, where behavior is specified in natural language and neutral logic before implementation in C/C++).

Scope:
Defective requirements come in all types of variable difficulty across multiple avionics and mission systems subsystems

- Simple: incorrect numerical values, inequalities, single line corrections, or referencing a wrong object type, element, name, or tag.
- Complex: full requirement section/document rewrites based on the behavior that is currently implemented
- Tracing variable changes across dependent documents, including timing and refresh dependencies.

Result:

- 157 defects closed/resolved from an 879 item backlog
- 18% of accumulated requirements debt resolved

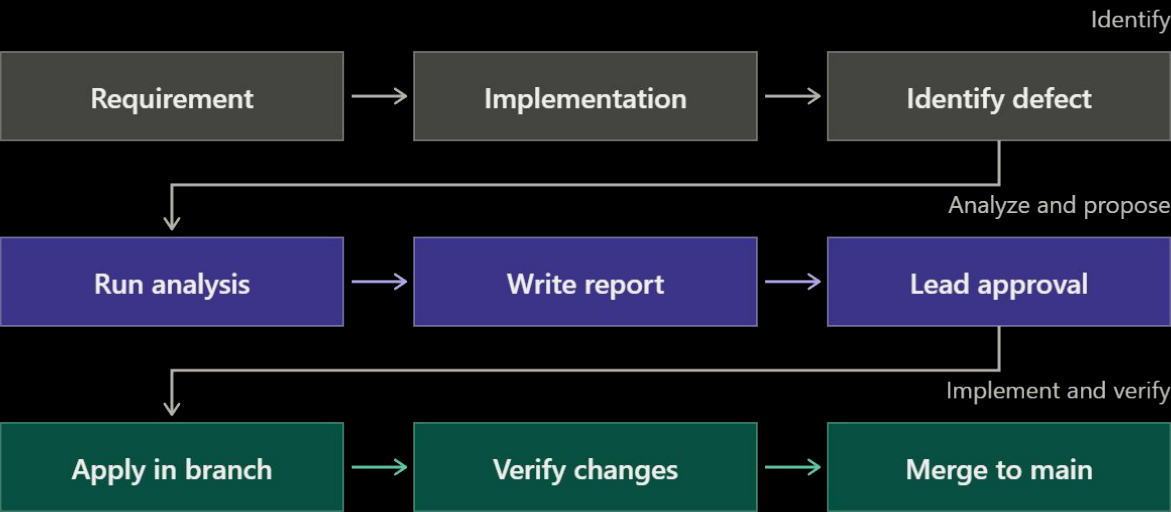
Involved Tools

JIRA, Git, GitLab, VS Code, IBM DOORS, Excel

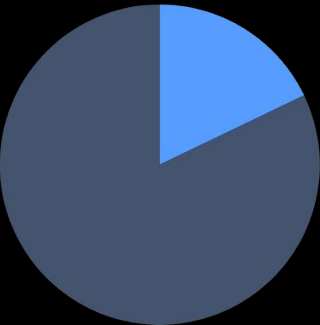


*Described at process level only, JSF program details are export controlled under ITAR

FEA Simulation with Ansys



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